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Beverage filling system and CIP processing method

Abstract

A beverage filling system for filling a container with a beverage includes: beverage supply piping that supplies the beverage; a beverage filling machine that is coupled to the beverage supply piping; CIP circulation piping that is coupled to the beverage filling machine, that feeds a cleaning liquid flowed out from the beverage filling machine during CIP processing toward the beverage supply piping, and that circulates the cleaning liquid; and a controller that controls the beverage filling system. A heater that heats the cleaning liquid that flows through the CIP circulation piping and a holding tube through which the cleaning liquid heated by the heater passes are provided in the CIP circulation piping. The cleaning liquid heated by the heater is set in such a way as to pass through the holding tube while spending a predetermined residence time or longer.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a beverage filling system and a CIP processing method.

BACKGROUND ART

(2) To date, it has been practiced that a large number of plastic bottles, which are being conveyed at a high speed, are continuously and aseptically filled with a content such as a carbonated beverage by using a filling machine such as a filler that is provided in a carbonated beverage aseptic filling apparatus. In such a carbonated beverage aseptic filling apparatus, a filling nozzle for filling the plastic bottle with the carbonated beverage is rotatably disposed in an aseptic chamber. (See, for example, PTLs 1 and 2).

CITATION LIST

Patent Literature

(3) PTL 1: Japanese Unexamined Patent Application Publication No. 2007-302325 PTL 2: Japanese Unexamined Patent Application Publication No. 2010-006429 PTL 3: International Publication No. 2014/098058

SUMMARY OF INVENTION

Technical Problem

(4) To date, regarding beverage supply piping of a beverage filling apparatus, CIP (Cleaning in Place) processing for cleaning the inside of the beverage supply piping is performed periodically or when switching the type of a product to be produced. Moreover, SIP (Sterilizing in Place) processing for sterilizing the inside of the beverage supply piping is performed. CIP processing is performed by causing a cleaning liquid in which, for example, an alkaline chemical agent such as sodium hydroxide is added to water to flow through a flow path extending from the inside of a pipe

line of the beverage supply piping to a filling nozzle of a filling machine, and then causing a cleaning liquid in which an acid chemical agent is added to water to flow through the flow path. SIP processing is processing for sterilizing the inside of the beverage supply piping before starting an operation of filling the product. In SIP processing, for example, sterilization processing at a high temperature is performed by causing heated water vapor or hot water to flow through the beverage supply piping that has been cleaned by the CIP processing.

(5) Against such a background, the inventors have found that the time required for SIP processing can be reduced by simultaneously performing CIP processing and SIP processing (PTL 3).

(6) The present disclosure provides a beverage filling system and a CIP processing method with which it is possible to simultaneously perform CIP processing and SIP processing by using low-cost facilities.

Solution to Problem

(7) A carbonated beverage filling system according to one embodiment is a beverage filling system for filling a container with a beverage including: beverage supply piping that supplies the beverage; a beverage filling machine that is coupled to the beverage supply piping; CIP circulation piping that is coupled to the beverage filling machine, that feeds a cleaning liquid flowed out from the beverage filling machine during CIP processing toward the beverage supply piping, and that circulates the cleaning liquid; and a controller that controls the beverage filling system. A heater that heats the cleaning liquid that flows through the CIP circulation piping and a holding tube through which the cleaning liquid heated by the heater passes are provided in the CIP circulation piping. The cleaning liquid heated by the heater is set in such a way as to pass through the holding tube while spending a predetermined residence time or longer.

(8) In the carbonated beverage filling system according to one embodiment, a thermometer may be provided on an outlet side of the holding tube of the CIP circulation piping, and the controller may monitor an F-value calculated based on a temperature of the thermometer.

(9) In the carbonated beverage filling system according to one embodiment, a time required by the cleaning liquid to pass through the holding tube may be predetermined based on an F-value required to sterilize the beverage.

(10) In the carbonated beverage filling system according to one embodiment, the cleaning liquid may be heated to 85° C. or higher and lower than 150° C. in the heater.

(11) In the carbonated beverage filling system according to one embodiment, an aseptic tank may be provided in the beverage supply piping, a bypass flow path that couples the CIP circulation piping on an outlet side of the holding tube and the beverage supply piping on an outlet side of the aseptic tank may be provided, and the bypass flow path may cause the cleaning liquid to flow from the holding tube side to the beverage supply piping on the outlet side of the aseptic tank without causing the cleaning liquid to pass through the aseptic tank.

(12) A CIP processing method according to one embodiment is a CIP processing method for performing CIP processing on a beverage filling system for filling a container with a beverage. The beverage filling system includes beverage supply piping that supplies the beverage, a beverage filling machine that is coupled to the beverage supply piping, and CIP circulation piping that is coupled to the beverage filling machine, that feeds a cleaning liquid flowed out from the beverage filling machine during CIP processing toward the beverage supply piping, and that circulates the cleaning liquid. The CIP processing method includes: a step of heating the cleaning liquid that flows through the CIP circulation piping by using a heater; and a step of causing the cleaning liquid heated by the heater to pass through the holding tube. The cleaning liquid heated by the heater is set in such a way as to pass through the holding tube while spending a predetermined residence time or longer.

(13) According to the present disclosure, it is possible to perform CIP processing and SIP processing simultaneously by using low-cost facilities.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a schematic view illustrating the configuration of a beverage filling system according to one embodiment.
- (2) FIG. 2 is a schematic view illustrating the flow of a fluid in a beverage filling machine of the beverage filling system according to one embodiment and in the vicinity thereof.
- (3) FIG. 3 is a schematic sectional view illustrating a filling nozzle of the beverage filling machine of the beverage filling system according to one embodiment.
- (4) FIG. 4 is a schematic view of flow paths of the beverage filling system to be CIP cleaned.
- (5) FIG. 5 is a schematic sectional view illustrating flow paths to be CIP cleaned during first CIP processing in the filling nozzle.
- (6) FIG. 6 is a schematic sectional view illustrating flow paths to be CIP cleaned during second CIP processing in the filling nozzle.
- (7) FIG. 7 is a schematic sectional view illustrating flow paths to be CIP cleaned during third CIP processing in the filling nozzle.

DESCRIPTION OF EMBODIMENTS

- (8) Hereafter, one embodiment will be described with reference to FIGS. 1 to 7. FIGS. 1 to 7 are views illustrating one embodiment. In the following figures, the same portions will be denoted by the same numerals, and detailed descriptions may be partially omitted.
- (9) (Beverage Filling System)
- (10) First, referring to FIGS. 1 and 2, the configuration of the entirety of a beverage filling system according to the present embodiment will be described.
- (11) A beverage filling system (aseptic filling system) **10** illustrated in FIG. 1 is a system that is used for both of a carbonated beverage and a non-carbonated beverage. That is, the beverage filling system **10** is an aseptic filling system that can fill a bottle (container) **30** (see FIG. 2) selectively with a beverage including a carbonated beverage and a beverage including a non-carbonated beverage. The bottle **30** can be made by biaxial-stretch blow molding a preform made by injection molding a synthetic resin material. It is preferable that a thermoplastic resin, in particular, PE (polyethylene), PP (polypropylene), PET (polyethylene terephthalate), or PEN (polyethylene naphthalate) be used as the material of the bottle **30**. Alternatively, the container may be a paper container, a glass bottle, a can, or the like. The container may be a compound container that is a combination of two or more of a plastic container, a paper container, a glass bottle, a can, and the like. In the present embodiment, an example in which a plastic bottle is used as the container will be described.
- (12) As illustrated in FIG. 1, the beverage filling system **10** includes a beverage sterilizing device **41**, an aseptic tank **42**, a carbonated beverage generating unit **44**, and a beverage filling machine (filler) **20**.
- (13) Among these, the beverage sterilizing device **41** sterilizes, for example, a material liquid including a component derived from animals and plants such as a fruit juice or a milk component. The beverage sterilizing device **41** may be, for example, an ultra-high-temperature (UHT) instantaneous sterilizing device.
- (14) The aseptic tank **42** temporarily stores a sterilized beverage sterilized by the beverage sterilizing device **41**. It is not necessary to provide the aseptic tank **42**, and the sterilized beverage from the beverage sterilizing device **41** may be directly supplied to the carbonated beverage generating unit **44** or a beverage filling tank **75**.
- (15) A first pump **51** is provided between the aseptic tank **42** and the beverage filling tank **75**. The first pump **51** feeds a liquid such as a beverage from the aseptic tank **42** toward the beverage filling tank **75**. Without providing the first pump **51**, a liquid such as a beverage may be fed toward the

beverage filling tank **75** by using a pressure from the aseptic tank **42**.

(16) The carbonated beverage generating unit **44** is used when filling a container with a carbonated beverage by using the beverage filling machine **20**. The carbonated beverage generating unit **44** produces an aseptic carbonated beverage by injecting carbon dioxide gas into a beverage from the beverage sterilizing device **41** and thereby dissolving the carbon dioxide gas in the beverage. The carbonated beverage generating unit **44** may include, for example, a beverage cooling device and a carbonator.

(17) The beverage sterilizing device **41**, the aseptic tank **42**, the carbonated beverage generating unit **44**, and the beverage filling machine **20** are coupled by beverage supply piping **65**. The beverage supply piping **65** is piping that supplies a beverage to the beverage filling machine **20**, and the beverage sequentially passes through the inside of the beverage supply piping **65**. When CIP processing described below is performed, a cleaning liquid passes through the inside of the beverage supply piping **65**.

(18) CIP circulation piping **81** is coupled to the beverage filling machine **20**. The CIP circulation piping **81** is a line that feeds a cleaning liquid flowed out from the beverage filling machine **20** during CIP processing toward the beverage supply piping **65** and that circulates the cleaning liquid. The CIP circulation piping **81** couples the beverage filling machine **20** and a middle part of the beverage supply piping **65**. In the CIP circulation piping **81**, a second pump **52**, a heat exchanger **61**, a CIP tank **85**, a third pump **91**, a heater **93**, and a holding tube **62** are provided in order from the beverage filling machine **20** side.

(19) The second pump **52** feeds the cleaning liquid from the beverage filling machine **20** toward the CIP tank **85** or toward an outlet flow path **61b** of the heat exchanger **61** described below.

(20) The heat exchanger **61** is provided between the second pump **52** and the CIP tank **85**. The heat exchanger **61** has an inlet flow path **61a**, into which a liquid such as aseptic water flows during cleaning of the beverage filling system **10**, and the outlet flow path **61b**, through which a liquid waste from the beverage filling machine **20** flows out. The temperature of the liquid such as aseptic water supplied from the inlet flow path **61a** rises while exchanging heat inside of the heat exchanger **61** with a high-temperature liquid waste from the beverage filling machine **20**. Thus, it is possible to reduce energy required by the heater **93** to increase the temperature of a cleaning liquid including aseptic water and the like. When the cleaning liquid is circulated in the CIP circulation piping **81**, the cleaning liquid flows through a heat exchanger bypass flow path **61c** that bypasses the heat exchanger **61**.

(21) The CIP tank **85** temporarily stores a cleaning liquid from the beverage filling machine **20**. A cleaning liquid supply source **63**, which supplies an alkaline cleaning liquid to the CIP circulation piping **81**, is connected between the CIP tank **85** and the third pump **91**. The third pump **91** feeds the cleaning liquid from the CIP tank **85** toward the heater **93**. Instead of an alkaline cleaning liquid, the cleaning liquid supply source **63** may supply another cleaning liquid such as an acid cleaning liquid, a deodorant, or the like.

(22) The heater **93** heats a cleaning liquid that flows through the CIP circulation piping **81**. For example, a plate heat exchanger or a shell-and-tube heat exchanger can be used as the heater **93**. The heater **93** heats the cleaning liquid to, for example, 80° C. or higher and 150° or lower, or 85° C. or higher and 100° C. or lower, preferably 90° C. or higher and lower than 100° C., and more preferably 95° C. or higher and lower than 100° C.

(23) The holding tube **62** is provided in the CIP circulation piping **81** between the heater **93** and a connection portion where the CIP circulation piping **81** and the beverage supply piping **65** are connected. The holding tube **62** includes a coil-shaped curved tube, a straight tube, a spiral tube, and the like, and performs heat processing or sterilization processing while a cleaning liquid flows therethrough. The cleaning liquid is set in such a way as to pass through the holding tube **62** while spending a predetermined residence time or longer. In this way, the cleaning liquid resides in the holding tube **62** for a certain residence time (holding time) while maintaining a sterilization

temperature, and thus the sterility of the cleaning liquid can be ensured.

(24) A bypass flow path **66** is provided between the holding tube **62** and the first pump **51**. The bypass flow path **66** couples the CIP circulation piping **81** on the outlet side of the holding tube **62** (on the inlet side of the aseptic tank **42**) and the beverage supply piping **65** on the outlet side of the aseptic tank **42**. The bypass flow path **66** causes the cleaning liquid to flow from the holding tube **62** side to the beverage supply piping **65** on the outlet side of the aseptic tank **42** without causing the cleaning liquid to pass through the aseptic tank **42**. Thus, it is possible to clean and sterilize the aseptic tank **42** independently from other elements of the beverage supply piping **65**. For example, during CIP processing described below, while causing a heated cleaning liquid to flow from the holding tube **62** side to the first pump **51** side via the bypass flow path **66**, it is possible to perform the CIP processing on the aseptic tank **42** by using another cleaning liquid that has passed through the beverage sterilizing device **41**.

(25) Thermometers **68a** to **68d** and a flowmeter **69** are disposed in the beverage supply piping **65** and the CIP circulation piping **81**. The thermometers **68a** to **68d** each measure the temperature of a liquid that flows in each piping. The flowmeter **69** measures the flow rate of the liquid that flows in each piping. To be specific, in the CIP circulation piping **81**, the thermometer **68a** and the flowmeter **69** are disposed on the outlet side of the heater **93**, and a thermometer **68b** is disposed on the outlet side of the holding tube **62**. In the beverage supply piping **65**, the thermometer **68c** is disposed on the outlet side of the beverage filling machine **20**. Moreover, the thermometer **68d** is disposed in the bypass flow path **66**.

(26) A controller **60** controls all or part of the beverage filling system **10**. The controller **60** may include a plurality of controllers that independently control the elements of the beverage filling system **10**.

(27) As illustrated in FIG. 2, the beverage filling machine **20** fills the bottle **30** from the mouth of the bottle **30** with an aseptic carbonated beverage or an aseptic non-carbonated beverage that has been sterilized beforehand or an unsterilized carbonated beverage that does not need to be sterilized (hereafter, simply referred to as “beverage”). In the beverage filling machine **20**, the bottle **30** in an empty state is filled with the beverage. In the beverage filling machine **20**, while a plurality of bottles **30** are rotated (revolved), the inside of each of the bottles **30** is filled with the beverage.

(28) In a case where the beverage with which the bottle **30** is to be filled is a carbonated beverage (an aseptic carbonated beverage or an unsterilized carbonated beverage), the bottle **30** is filled with the carbonated beverage at a filling temperature of 1° C. or higher and 40° C. or lower, and preferably 5° C. or higher and 10° C. or lower. In this way, the filling temperature of the carbonated beverage is set at, for example, 1° C. or higher and 10° C. or lower, because carbon dioxide gas tends to escape from the carbonated beverage if the liquid temperature of the carbonated beverage becomes higher than 10° C. Examples of a carbonated beverage with which a container is to be filled by the beverage filling machine **20** include various beverages including carbon dioxide gas, which are, for example, carbonated soft drinks such as cider, cola, and the like, and alcohol drinks such as beer.

(29) In a case where the drink with which the bottle **30** is to be filled is an aseptic non-carbonated beverage, the bottle **30** is filled with the non-carbonated beverage at a filling temperature of 1° C. or higher and 40° C. or lower, and preferably 10° C. or higher and 30° C. or lower. Examples of an aseptic non-carbonated beverage with which a container is to be filled by the beverage filling machine **20** include a non-carbonated beverage including a component derived from animals and plants, such as a fruit juice or a milk component, and mineral water or the like that does not include a component derived from animals and plants.

(30) The beverage filling system **10** has an aseptic chamber **13** whose inside is maintained in an aseptic state. The beverage filling machine **20** is provided in the aseptic chamber **13**. The beverage filling tank (a filling head tank, a buffer tank) **75** is disposed at a position that is outside of the aseptic chamber **13** and above the beverage filling machine **20**. The inside of the beverage filling

tank **75** is filled with a beverage. The pressure **P1** of the inside of the beverage filling tank **75** is measured by a first pressure gauge **64** provided on the beverage filling tank **75**. The beverage filling tank **75** need not be set above the beverage filling machine **20**, and may be set on a floor on which the beverage filling machine **20** is set.

(31) The beverage supply piping **65** described above is coupled to the beverage filling tank **75**. As illustrated in FIG. **1**, the CIP circulation piping **81** is coupled to the beverage supply piping **65**.

(32) A beverage supply line **73** is coupled to the beverage filling tank **75**. The beverage supply line **73** supplies a beverage with which the beverage filling tank **75** is filled toward a filling nozzle **72** described below. The beverage filling tank **75** is coupled to the filling nozzle **72** via the beverage supply line **73**.

(33) Moreover, a counter gas line **74** is coupled to the beverage filling tank **75**. The counter gas line **74** is used in a case where a beverage with which a container is to be filled is a carbonated beverage, and supplies aseptic carbon dioxide gas with which the beverage filling tank **75** is filled toward the filling nozzle **72** described below. The beverage filling tank **75** is coupled to the filling nozzle **72** via the counter gas line **74**.

(34) A counter gas valve **67** is provided at a connection portion between the beverage filling tank **75** and the counter gas line **74**. The counter gas valve **67** is directly connected to the beverage filling tank **75**. The counter gas valve **67** is opened when a beverage with which a container is to be filled is a carbonated beverage, and is closed when the beverage is a non-carbonated beverage.

(35) In the beverage filling machine **20**, the bottle **30** in an empty state is filled with a beverage with which the beverage filling tank **75** has been filled. The beverage filling machine **20** has a conveyance wheel **71** that rotates around an axis parallel to the vertical direction. While a plurality of bottles **30** are rotated (revolved) by the conveyance wheel **71**, the inside of each of the bottles **30** is filled with the beverage. A plurality of filling nozzles **72** are arranged along the outer periphery of the conveyance wheel **71**. One bottle **30** is attached to each filling nozzle **72**, and the beverage is injected from the filling nozzle **72** to the inside of the bottle **30**. The configuration of the filling nozzle **72** will be described below.

(36) The conveyance wheel **71**, the filling nozzle **72**, at least a part of the beverage supply line **73**, and at least a part of the counter gas line **74** are surrounded by a cover **76** that constitutes a part of the aseptic chamber **13**. A rotary joint **77** is attached to an upper part of the cover **76**. The beverage supply line **73** and the counter gas line **74** are attached to the cover **76** of the aseptic chamber **13** by the rotary joint **77**. The rotary joint **77** aseptically seals rotational bodies (the conveyance wheel **71**, the filling nozzle **72**, and rotary piping and the like of the beverage supply line **73** and the counter gas line **74**) and non-rotational bodies (the cover **76**, and fixed piping and the like of the beverage supply line **73** and the counter gas line **74**).

(37) The beverage supply line **73** and the counter gas line **74** are coupled to each filling nozzle **72**. During filling with a beverage, one end of the beverage supply line **73** is coupled to the beverage filling tank **75** filled with the beverage and the other end of the beverage supply line **73** communicates with the inside of the bottle **30**. The beverage supplied from the beverage filling tank **75** passes through the beverage supply line **73**, and is injected to the inside of the bottle **30**.

(38) During filling with a beverage, one end of the counter gas line **74** is coupled to the beverage filling tank **75** and the other end of the counter gas line **74** communicates with the inside of the bottle **30**. A counter pressure gas, which is aseptic carbon dioxide gas supplied from the beverage filling tank **75**, passes through the counter gas line **74**, and the inside of the bottle **30** is filled with the counter pressure gas. A counter manifold (counter gas branching portion) **53** is provided at a middle part of the counter gas line **74**. The counter gas line **74** from the beverage filling tank **75** branches at the counter manifold **53** into a plurality of lines each of which extends to a corresponding filling nozzle **72**.

(39) Moreover, a snift line **78** is coupled to each filling nozzle **72**. The snift line **78** is used when a beverage with which a container is to be filled is a carbonated beverage. One end of the snift line

78 is coupled to the counter gas line **74**, and the other end of the snift line **78** extends to the outside of the aseptic chamber **13**. A gas inside of the bottle **30** can be discharged via the snift line **78**. A snift manifold (snift line branching portion) **56** is provided at a middle part of the snift line **78**. Carbon dioxide gas from the snift line **78** is gathered in the snift manifold **56** and discharged into the aseptic chamber **13**. A discharge valve **79** is provided in the snift line **78** in the aseptic chamber **13**. Due to the discharge valve **79**, carbon dioxide gas from the snift line **78** is discharged into the aseptic chamber **13**. In this way, carbon dioxide gas from the snift line **78** is discharged into the aseptic chamber **13** by using the discharge valve **79**. Thus, carbon dioxide gas in the bottle **30** can be discharged into the aseptic chamber **13**, which is an aseptic space, without allowing contamination with bacteria. The snift manifold **56** and the counter manifold **53** are coupled by a first bypass line **54**. A first valve **55** is provided in the first bypass line **54**, and the first valve **55** is normally closed. Without providing the discharge valve **79** in the snift line **78**, the snift line **78** may be connected to the rotary joint **77** and carbon dioxide gas may be discharged from the rotary joint **77** to the outside of the aseptic chamber **13**. In the case illustrated in the figures, the rotary joint **77** is provided at an upper part of the beverage filling machine **20**. This is not a limitation, and the rotary joint **77** may be set at a lower part of the beverage filling machine **20**. A rotary joint may be provided at each of an upper part and a lower part of the beverage filling machine **20**.

(40) It is preferable that a flow path, in the beverage filling system **10**, through which a beverage passes be performed CIP (Cleaning in Place) processing periodically or when switching the type of beverage. CIP processing is performed by causing an acid cleaning liquid to flow through a flow path after causing an alkaline cleaning liquid to flow through the flow path or before causing the alkaline cleaning liquid to flow through the flow path. An alkaline cleaning liquid is water to which an alkaline chemical agent that is a mixture of caustic soda (sodium hydroxide), potassium hydroxide, sodium carbonate, sodium silicate, sodium phosphate, sodium hypochlorite, a surface-active agent, chelating agent, and the like is added. An acid cleaning liquid is water to which an acid chemical agent such as a nitric-acid agent or a phosphoric-acid agent is added. An alkaline cleaning step using the alkaline cleaning liquid and an acid cleaning step using the acid cleaning liquid may be freely combined and performed. Thus, residues of a previous beverage and the like, which have adhered to the inside of flow paths through which a beverage is to pass, are removed. Optionally, SIP (Sterilizing in Place) processing may be performed. SIP processing is processing that is performed before starting an operation of filling a container with a beverage in order to sterilize the inside of a flow path through which the beverage is to pass, and is performed by, for example, causing heat steam or hot water to flow through a flow path that has been cleaned by the CIP processing described above. Thus, the flow path through which the beverage is to pass is sterilized to be in an aseptic state.

(41) In order to perform CIP processing described above, a CIP cup **82** that receives a cleaning liquid from the filling nozzle **72** is provided in the vicinity of the filling nozzle **72**. A CIP line **83** is coupled to the CIP cup **82**. One end of the CIP line **83** is coupled to the CIP cup **82**, and the other end of the CIP line **83** is coupled to the CIP tank **85** disposed outside of the aseptic chamber **13**. It is possible to discharge a cleaning liquid from the filling nozzle **72** to the CIP tank **85** via the CIP line **83**. The CIP line **83** is coupled to a CIP manifold (CIP line branching portion) **59**, and the CIP manifold **59** is coupled to the CIP circulation piping **81**. A cleaning liquid from the CIP line **83** is collectively recovered in the CIP manifold **59** and discharged to the CIP tank **85** via the CIP circulation piping **81**. The CIP manifold **59** and the snift manifold **56** are coupled by a second bypass line **57**. A second valve **58** is provided in the second bypass line **57**. The second valve **58** is normally closed.

(42) An exhaust line **89** that discharges a gas inside of the CIP tank **85** is provided on an upper part of the CIP tank **85**. A scrubber (not shown) that treats the gas is coupled to the exhaust line **89**.

(43) An aseptic air supply device **70** that feeds a large volume of aseptic air into the aseptic chamber **13** is provided on the cover **76** of the aseptic chamber **13**. The aseptic air supply device **70**

introduces aseptic air into the aseptic chamber **13**. Thus, the inside of the aseptic chamber **13** and the aseptic area of the beverage filling machine **20** are all maintained at a positive pressure, and entry of external air into the aseptic chamber **13** is suppressed. A large volume of aseptic air is fed into the aseptic chamber **13** by the aseptic air supply device **70**. Thus, even in a case where carbon dioxide gas is discharged into the aseptic chamber **13** from the discharge valve **79** as described above, the carbon dioxide gas concentration in the aseptic chamber **13** is not likely to increase excessively. The supply rate of aseptic air for achieving the above object is 5 m.sup.3/min or greater and 100 m.sup.3/min or less, and preferably 10 m.sup.3/min or greater and 50 m.sup.3/min or less.

(44) (Filling Nozzle)

(45) Next, referring to FIG. **3**, the configuration of the filling nozzle **72** of the beverage filling machine **20** will be described. FIG. **3** illustrates the filling nozzle **72** during CIP processing, and the CIP cup **82** is disposed below the filling nozzle **72**.

(46) As illustrated in FIG. **3**, the filling nozzle **72** has a body portion **72a**. The beverage supply line **73** and the counter gas line **74** are each coupled to the body portion **72a**. The beverage supply line **73** and the counter gas line **74** extend through the rotary joint **77** provided on the cover **76**.

(47) An upper end of the beverage supply line **73** is coupled to the beverage filling tank **75**, and a lower end of the beverage supply line **73** is open toward the CIP cup **82**. A cleaning liquid supplied from the beverage filling tank **75** passes through the beverage supply line **73** and flows to the inside of the CIP cup **82**. The cleaning liquid flowed to the inside of the CIP cup **82** flows into the CIP manifold **59** via the CIP line **83**. Subsequently, the cleaning liquid is discharged from the CIP manifold **59** to the outside of the beverage filling machine **20**.

(48) The counter gas line **74** is used when a beverage with which a container is to be filled is a carbonated beverage. An upper end of the counter gas line **74** is coupled to the beverage filling tank **75**, and a lower end of the counter gas line **74** is open toward the CIP cup **82**. The snift line **78** is coupled a middle part of the counter gas line **74**. A cleaning liquid supplied from the beverage filling tank **75** passes through the counter gas line **74** and flows to the inside of the CIP cup **82**. Alternatively, a cleaning liquid supplied from the beverage filling tank **75** flows into the snift manifold **56** via the snift line **78**. Subsequently, the cleaning liquid from the snift manifold **56** passes through the snift line **78**, and then is discharged into the aseptic chamber **13** from the discharge valve **79**. The snift line **78** may discharge the cleaning liquid to the outside of the beverage filling machine **20** (not shown) from the rotary joint **77** positioned at an upper part of the beverage filling machine **20**. A rotary joint may be additionally provided at a lower part of the beverage filling machine **20** to discharge the cleaning liquid from the snift line **78** to the outside of the beverage filling machine **20** (not shown).

(49) (Aseptic Carbonated Beverage Filling Method)

(50) Next, an aseptic carbonated beverage filling method using the beverage filling system **10** will be described. Hereafter, an aseptic carbonated beverage filling method in normal time, that is, an aseptic carbonated beverage filling method for filling the bottle **30** with an aseptic carbonated beverage to produce a product bottle will be described.

(51) First, an empty bottle **30** that has been sterilized is conveyed to the beverage filling machine **20**. In the beverage filling machine **20**, while the bottle **30** is rotated (revolved), the bottle **30** is filled with an aseptic carbonated beverage from the mouth of the bottle **30**. In the beverage filling machine **20**, the sterilized bottle **30** is filled with the aseptic carbonated beverage fed from the beverage filling tank **75** at a filling temperature of 1° C. or higher and 40° C. or lower, and preferably 5° C. or higher and 10° C. or lower.

(52) During this time, in the beverage filling machine **20**, the filling nozzle **72** is in close contact with the mouth of the bottle **30**, and the counter gas line **74** and the bottle **30** communicate with each other. At this time, the snift line **78** is closed. Next, aseptic carbon dioxide gas for counter pressure is supplied from the beverage filling tank **75** to the inside of the bottle **30** via the counter

gas line **74**. Thus, the internal pressure of the bottle **30** is increased to a level higher than the atmospheric pressure, and the internal pressure of the bottle **30** becomes the same as the internal pressure of the beverage filling tank **75**.

(53) Next, the inside of the bottle **30** is filled with the aseptic carbonated beverage from the beverage supply line **73**. In this case, the aseptic carbonated beverage from the beverage filling tank **75** passes through the beverage supply line **73**, and is injected to the inside of the bottle **30**.

(54) Subsequently, supply of the aseptic carbonated beverage from the beverage supply line **73** is stopped. Next, the beverage supply line **73** and the counter gas line **74** are closed, the snift line **78** is opened, and the gas inside of the bottle **30** is discharged from the snift line **78**. Thus, the internal pressure of the bottle **30** becomes the same as the atmospheric pressure, and filling of the bottle **30** with the aseptic carbonated beverage is finished. At this time, the gas from the bottle **30** passes through the snift line **78**, and then is discharged from the discharge valve **79** into the aseptic chamber **13**. Next, the filling nozzle **72** separates from the mouth of the bottle **30**, and the bottle **30** is conveyed to a capper (not shown).

(55) Subsequently, a cap (not shown) is put on the bottle **30**, which has been filled with the aseptic carbonated beverage by the beverage filling machine **20**, thereby obtaining a product bottle.

(56) It is preferable that the production (conveyance) speed of the bottle **30** in the beverage filling system **10** be 100 bpm or faster and 1500 bpm or slower. Here, the term “bpm (bottle per minute)” refers to the conveyance speed of the bottle **30** per one minute.

(57) (Aseptic Non-Carbonated Beverage Filling Method)

(58) Next, an aseptic non-carbonated beverage filling method using the beverage filling system **10** will be described. Hereafter, an aseptic non-carbonated beverage filling method in normal time, that is, an aseptic non-carbonated beverage filling method for filling the bottle **30** with a non-aseptic carbonated beverage to produce a product bottle will be described.

(59) First, an empty bottle **30** that has been sterilized is conveyed to the beverage filling machine **20**. Next, in the beverage filling machine **20**, in a state in which the filling nozzle **72** is not in close contact with the mouth of the bottle **30**, the inside of the bottle **30** is filled with an aseptic non-carbonated beverage from the beverage supply line **73**. The aseptic non-carbonated beverage from the beverage filling tank **75** passes through the beverage supply line **73** and is injected to the inside of the bottle **30**. Subsequently, supply of the aseptic non-carbonated beverage from the beverage supply line **73** is stopped. At this time, the counter gas line **74** and the snift line **78** are respectively closed by the counter gas valve **67** and another valve (not shown).

(60) A cap (not shown) is put on the bottle **30**, which has been filled with the non-aseptic carbonated beverage by the beverage filling machine **20**, thereby obtaining a product bottle.

(61) (CIP Processing Method)

(62) Next, an operation of CIP (Cleaning in Place) processing that is performed in the beverage filling system **10**, for example, periodically or when switching the type of beverage will be described. Control of CIP processing described below is controlled by the controller **60**.

(63) First, water is fed into the CIP circulation piping **81** from the inlet flow path **61a** of the heat exchanger **61**. As the water circulates, the inside of the CIP circulation piping **81**, the inside of the beverage supply piping **65**, and the inside of the beverage filling machine **20** are each cleaned.

(64) Next, as illustrated in FIG. **4**, an alkaline cleaning liquid is fed from the cleaning liquid supply source **63**. As the alkaline cleaning liquid circulates, the inside of the CIP circulation piping **81**, the inside of the beverage supply piping **65**, and the inside of the beverage filling machine **20** are cleaned. In FIG. **4**, flow paths through which the alkaline cleaning liquid passes are represented by thick lines and hatchings.

(65) During this time, the alkaline cleaning liquid is fed into the heater **93** by the third pump **91** located in the CIP circulation piping **81**. The alkaline cleaning liquid is heated in the heater **93** to, for example, 85°C or higher and 100° C. or lower, preferably 90° C. or higher and lower than 100° C., and more preferably 95° C. or higher and lower than 100° C. The heated alkaline cleaning

liquid reaches the beverage supply piping **65** via the holding tube **62**. Next, the heated alkaline cleaning liquid reaches the beverage filling machine **20** sequentially via the aseptic tank **42**, the first pump **51**, and the beverage filling tank **75**. Subsequently, the alkaline cleaning liquid flows out from the beverage filling machine **20** to the CIP circulation piping **81**, and is fed to the heater **93** again sequentially via the second pump **52**, the CIP tank **85**, and the third pump **91**. In this way, after circulating the alkaline cleaning liquid to clean the inside of the CIP circulation piping **81**, the inside of the beverage supply piping **65**, and the inside of the beverage filling machine **20** for a predetermined time, the alkaline cleaning liquid is discharged to the outside from the outlet flow path **61b** of the heat exchanger **61**.

(66) In a case where a liquid including sodium hydroxide or potassium hydroxide by 0.1 mass % or more and 10 mass % or less is used as the alkaline cleaning liquid, the alkaline cleaning liquid is heated to the aforementioned temperature by the heater **93** provided in the CIP circulation piping **81**. The heated alkaline cleaning liquid is supplied to each of the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20**. When the circulation is performed for, for example, about 5 minutes or longer and 60 minutes or shorter, the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20** are each appropriately cleaned. At the same time, the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20** are each sterilized, and thus SIP processing is simultaneously performed without additionally performing SIP processing (CSIP processing). In this way, sterilization is performed simultaneously with CIP processing for cleaning various devices of the beverage filling system **10**. Thus, the time required to perform SIP processing can be reduced, or SIP processing itself can be omitted. Thus, it is possible to reduce the product switching time of the beverage filling system **10** and to increase the production capacity.

(67) Next, in a similar way, an acid cleaning liquid is caused to flow through the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20** to acid-clean the entirety of the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20**. Subsequently, aseptic water is caused to flow through all of the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20** to rinse the entirety of the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20**. In this way, residues of a previous beverage and the like, which have adhered to the inside of flow paths through which a beverage is to pass, are removed. The acid cleaning liquid is heated in the heater **93** provided in the CIP circulation piping **81** to, for example, 85° C. or higher and 100° C. or lower, preferably 90° C. or higher and lower than 100° C., and more preferably 95° C. or higher and lower than 100° C. The heated acid cleaning liquid is supplied to each of the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20**. When the circulation is performed, for example, 5 minutes or longer and 30 minutes or shorter, the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20** are each appropriately cleaned. At the same time, the CIP circulation piping **81**, the beverage supply piping **65**, and the beverage filling machine **20** are each sterilized, and thus SIP processing is performed without additionally performing SIP processing (CSIP processing). The order of using the acid cleaning liquid and using the alkaline cleaning liquid may be determined as appropriate in view of cleaning performance, and, for example, acid cleaning may be performed first and alkaline cleaning may be performed next. Alternatively, only alkaline cleaning may be performed, or only acid cleaning may be performed.

(68) After finishing CIP processing, the cleaning liquid used for the CIP processing is discharged from the CIP circulation piping **81**, and the cleaning liquid that remains in the beverage supply piping **65** and the CIP circulation piping **81** is washed away by using aseptic water. At the time when the cleaning liquid in the beverage supply piping **65** and the CIP circulation piping **81** is removed by using the aseptic water and all cleaning liquid in the filling nozzle **72** of the beverage filling machine **20** is replaced with aseptic water, feeding of aseptic water to the beverage supply

piping **65** and the CIP circulation piping **81** is stopped. Simultaneously with or subsequently to this, while removing aseptic water that remains in the aseptic tank **42** and the beverage filling tank **75**, aseptic air is supplied into the beverage supply piping **65** including the aseptic tank **42** and the beverage filling tank **75**. Thus, the inside of each of the aseptic tank **42**, the beverage filling tank **75**, the beverage supply piping **65**, and the CIP circulation piping **81**, which have been performed the CIP processing, is maintained at a positive pressure to maintain sterility. Subsequently, while maintaining the positive pressure, aseptic water that has accumulated in the aseptic tank **42**, the beverage supply piping **65**, the beverage filling tank **75**, and the beverage filling machine **20** may be air-blown to remove the aseptic water from a drain line (not shown) provided at each place. Thus, it is possible to reduce the risk that a beverage with which a container is to be filled might be diluted at the start of production.

(69) After rinsing has finished, a beverage is stored in the aseptic tank **42**, next the beverage passes through the beverage supply piping **65** and reaches the beverage filling machine **20**, and a production step of performing an operation of filling the bottle **30** with the beverage is started.

(70) (CIP Cleaning Liquid Heating Method)

(71) Next, a CIP cleaning liquid heating method for heating an alkaline cleaning liquid or an acid cleaning liquid (hereafter, referred to as “CIP cleaning liquid”) during the CIP processing will be described.

(72) As described above, the CIP cleaning liquid is fed into the heater **93** of the CIP circulation piping **81** and heated in the heater **93** to, for example, 85° C. or higher and lower than 100° C., preferably 90° C. or higher and lower than 100° C., and more preferably 95° C. or higher and lower than 100° C. The heated CIP cleaning liquid is supplied to the beverage supply piping **65** via the holding tube **62**. The CIP cleaning liquid requires a certain time (residence time) or longer to pass through the holding tube **62**, and maintains a predetermined temperature or higher during this time.

(73) The degree of sterilization of the cleaning liquid that passes through the holding tube **62** may be controlled by using an F-value. For example, while causing the CIP cleaning liquid to flow through the holding tube **62**, the temperature of the CIP cleaning liquid may be measured by using the thermometer **68b** disposed on the outlet side of the holding tube **62**. In this case, temperature information from the thermometer **68b** is sent to the controller **60** at certain time intervals. The controller **60** calculates an F-value at the time based on the temperature information from the thermometer **68b**. Here, the term “F-value” refers to a heating time required to kill all bacteria when the bacteria are heated for a certain time. The F-value, which is represented by the lethal time of bacteria, is calculated by using the following formula.

$$F = \int_{t_{sub.0}}^{t_{sub.1}} 10^{\frac{(T - T_r)}{Z}} dt \quad [\text{Math 1}]$$

(74) In the above formula, T is a temperature (° C.) measured by the thermometer **68b**, $10^{\frac{(T - T_r)}{Z}}$ is lethality at the sterilization temperature T, T_r is a reference temperature (° C.), and Z is a Z-value (° C.). t_{sub.1} (minutes) is the (shortest) residence time required by the CIP cleaning liquid to pass through the holding tube **62**, and is set beforehand as a predetermined value. Alternatively, t₁ (minutes) may be an actual passing time of the cleaning liquid that is a value measured in real time from the volume of the flowmeter **69** and the holding tube **62**.

(75) The controller **60** monitors the F-value calculated based on the temperature of the thermometer **68b** on the outlet side and continues CIP processing if this value maintains a predetermined value or greater. That is, the controller **60** integrates the value of $10^{\frac{(T - T_r)}{Z}}$ based on temperature information that is sent from the thermometer **68b** at certain time intervals. The integrated value in an interval between the present time and t_{sub.1} (minutes) immediately before the present time is determined as the F-value at the time. If the F-value maintains a predetermined value or greater, the controller **60** determines that the sterility of the CIP cleaning liquid that passes through the holding tube **62** is ensured and continues CIP processing. On the other hand, if the F-value becomes less than the predetermined value, the controller **60** may

determine that some trouble has occurred and the sterility of the CIP cleaning liquid is not ensured, and may stop CIP processing. Only if the F-value becomes less than the predetermined value, instead of supplying an unsterilized cleaning liquid to the beverage supply piping **65**, the unsterilized cleaning liquid may be drained from a blow valve (not shown). Subsequently, after the F-value has returned to the predetermined value, the cleaning liquid may be fed to the beverage supply piping **65**.

(76) As an example, if the pH of a beverage with which the bottle **30** is to be filled is, for example, 4 or higher and lower than 4.6, a sterilization temperature condition may be determined on the assumption that the reference temperature $T_r=85^{\circ}\text{C}$. and Z-value $=5^{\circ}\text{C}$. That is, the Food Sanitation Act stipulates that a bactericidal value necessary for sterilizing a beverage whose pH is 4 or higher and lower than 4.6 is equal to or greater than that of heating at 85°C . for 30 minutes ($F_{\text{sub.}85}\geq 30$). When Z $=5^{\circ}\text{C}$. is used, at 95°C ., it is possible to realize an equivalent bactericidal value by heating for 0.3 minutes (18 seconds). Therefore, the (shortest) residence time $t_{\text{sub.}1}$ (minutes) required by the CIP cleaning liquid to pass through the holding tube **62** is set to be 0.3 minutes (18 seconds), and if the temperature T of the thermometer **68b** on the outlet side maintains 95°C . or higher, it can be considered that the F-value maintains 30 or greater and the sterility of the CIP cleaning liquid is ensured. In cases where Z $=8^{\circ}\text{C}$. and 10°C . are used in order to further increase the sterilizing effect, the (shortest) residence time $t_{\text{sub.}1}$ (minutes) required by the CIP cleaning liquid to pass through the holding tube **62** may be respectively set to 1.7 minutes (101 seconds) and 3 minutes (180 seconds). In these cases, if the temperature T of the thermometer **68b** on the outlet side maintains 95°C . or higher, it is considered that the F-value maintains 30 or greater and the sterility of the CIP cleaning liquid is ensured. Thus, it is possible to supply the CIP cleaning liquid whose sterility is ensured to the beverage supply piping **65**.

(77) Moreover, in this case, it is not necessary to increase the temperature of the CIP cleaning liquid that passes through the CIP circulation piping **81** to higher than 100°C . Therefore, it is possible to handle each tank disposed in the CIP circulation piping **81** as a Class-2 pressure container stipulated in the Order for Enforcement of Industrial Safety and Health Act. Thus, various facilities necessary for CIP processing can be constructed at low cost, compared with a case where each tank disposed in the CIP circulation piping **81** is a Class-1 pressure container stipulated in the Order for Enforcement of Industrial Safety and Health Act. Note that, in order to perform CIP processing more efficiently, although the cost is high, each tank and the like may be changed to Class-1 pressure containers and CIP processing may be performed by using water of 100°C . or higher.

(78) In this way, it is possible to predetermine the (shortest) residence time $t_{\text{sub.}1}$ (minutes) required by the CIP cleaning liquid to pass through the holding tube **62** based on an F-value required for sterilization of a beverage, the Z-value, and the reference temperature T_r . For example, the residence time $t_{\text{sub.}1}$ is preferably 0.05 minutes or longer and 10 minutes or shorter, and more preferably 0.1 minutes or longer and 3 minutes or shorter. In order to allow each tank and the like disposed in the CIP circulation piping **81** to be a Class-2 pressure container, the reference temperature T_r is preferably lower than 100°C ., and more preferably 97°C . or lower. In order to prevent the residence time $t_{\text{sub.}1}$ from becoming unnecessarily long, the reference temperature T_r is preferably 87°C . or higher, and more preferably 90°C . or higher.

(79) In the above formula for calculating the F-value, the reference temperature T_r and the Z-value can be changed in accordance with the type of a beverage that is a product liquid. For example, when the pH of the product liquid is lower than 4, the reference temperature T_r may be 65°C . and the Z-value may be 5°C . That is, it is possible to appropriately change the values to be input to the calculation formula in accordance with the microorganism growth characteristics, the distribution temperature, and the like of the product liquid, which is a green tea beverage, mineral water, a chilled beverage, or the like.

(80) The sterilization method is not limited to the method in which the F-value is calculated as

described above, and, for example, a sterilization method in which temperature and time are used in a way known to date may be used.

(81) (Method for CIP Processing on Beverage Filling Machine)

(82) Next, a method for CIP processing on the beverage filling machine **20** during the CIP processing will be specifically described.

(83) In the present embodiment, regarding the beverage filling machine **20**, first CIP processing for CIP processing on a first piping channel and second CIP processing for CIP processing on a second piping channel are sequentially performed. Moreover, third CIP processing for CIP processing on a third piping channel may be performed. The first piping channel, the second piping channel, and the third piping channel, which are piping channels that differ from each other, may include a common flow path in a part thereof. The first piping channel, the second piping channel, and the third piping channel each may be a flow path through which a liquid flows or may be a flow path through which a gas flows during filling of a container with a beverage.

(84) The first piping channel is a piping channel in the beverage filling machine **20**, and includes at least the beverage supply line **73**. The second piping channel is a piping channel in the beverage filling machine **20**, and includes at least the counter gas line **74**. The third piping channel is a piping channel in the beverage filling machine **20**, and includes at least the snift line **78**.

(85) FIGS. **5** to **7** respectively illustrate flows of a CIP cleaning liquid when performing first CIP processing (first CIP processing step), second CIP processing (second CIP processing step), and third CIP processing (third CIP processing step). In FIGS. **5** to **7**, flow paths through which the CIP cleaning liquid passes are represented by thick lines, and flow paths through which the CIP cleaning liquid does not pass are represented by thin lines.

(86) As illustrated in FIG. **5**, during the first CIP processing, the CIP cleaning liquid flows from the beverage supply piping **65**, and flows into the beverage filling machine **20** via the beverage filling tank **75**. In the beverage filling machine **20**, the CIP cleaning liquid flows through the beverage supply line **73**, the filling nozzle **72**, the CIP cup **82**, the CIP line **83**, and the CIP manifold **59**, and flows out from the beverage filling machine **20**. Subsequently, the CIP cleaning liquid passes through the CIP circulation piping **81**, and flows into the CIP tank **85**. In this case, the first piping channel includes the beverage supply line **73**, the filling nozzle **72**, the CIP cup **82**, the CIP line **83**, and the CIP manifold **59** of the beverage filling machine **20**. During the first CIP processing, the CIP cleaning liquid does not flow through the counter gas line **74** and the snift line **78**. However, this is not a limitation, and the CIP cleaning liquid may flow through a part of the counter gas line **74** or a part of the snift line **78**.

(87) As illustrated in FIG. **6**, during the second CIP processing, the CIP cleaning liquid flows from the beverage supply piping **65**, and flows into the beverage filling machine **20** via the beverage filling tank **75**. In the beverage filling machine **20**, the CIP cleaning liquid flows through the counter gas line **74**, the counter manifold **53**, the filling nozzle **72**, the CIP cup **82**, the CIP line **83**, and the CIP manifold **59**, and flows out from the beverage filling machine **20**. Subsequently, the CIP cleaning liquid passes through the CIP circulation piping **81**, and flows into the CIP tank **85**. In this case, the second piping channel includes the counter manifold **53**, the counter gas line **74**, the filling nozzle **72**, the CIP cup **82**, the CIP line **83**, and the CIP manifold **59** of the beverage filling machine **20**. During the second CIP processing, the CIP cleaning liquid does not flow through the beverage supply line **73** and the snift line **78**. However, this is not a limitation, and the CIP cleaning liquid may flow through a part of the beverage supply line **73** or a part of the snift line **78**.

(88) As illustrated in FIG. **7**, during the third CIP processing, the CIP cleaning liquid flows from the beverage supply piping **65**, and flows into the beverage filling machine **20** via the beverage filling tank **75** and a part of the counter gas line **74**. In the beverage filling machine **20**, the CIP cleaning liquid flows through the counter manifold **53**, the first bypass line **54**, the snift manifold **56**, the snift line **78**, the filling nozzle **72**, the CIP cup **82**, the CIP line **83**, and the OP manifold **59**, and flows out from the beverage filling machine **20**. Subsequently, the CIP cleaning liquid passes

through the CIP circulation piping **81**, and flows into the CIP tank **85**. In the beverage filling machine **20**, the CIP cleaning liquid flows from the snift manifold **56**, and flows also into the CIP manifold **59** via the second bypass line **57**. In this case, the third piping channel includes the counter manifold **53**, the first bypass line **54**, the second bypass line **57**, the snift manifold **56**, the snift line **78**, the filling nozzle **72**, the CIP cup **82**, the CIP line **83**, and the CIP manifold **59** of the beverage filling machine **20**. During the second CIP processing, the CIP cleaning liquid does not flow through the beverage supply line **73**. However, this is not a limitation, and the CIP cleaning liquid may flow through a part of the beverage supply line **73**.

(89) The controller **60** switches between the first CIP processing, the second CIP processing, and the third CIP processing by appropriately on/off-controlling a valve (not shown) of each flow path. In this case, the first CIP processing, the second CIP processing, and the third CIP processing are respectively performed on all filling nozzles **72**. Thus, an advantageous effect that it is possible to completely sterilize all flow paths without changing flow paths, pumps, and the like can be obtained.

(90) The first CIP processing, the second CIP processing, and the third CIP processing may be performed in this order or may be performed in another order. The first CIP processing, the second CIP processing, and the third CIP processing may be performed for time periods that are the same as each other or may be performed for time periods that differ from each other. The second piping channel may include both of the counter gas line **74** and the snift line **78**. In this case, without performing the third CIP processing, the first CIP processing for performing CIP processing on the first piping channel and the second CIP processing for performing CIP processing on the second piping channel including the counter gas line **74** and the snift line **78** may be performed.

(91) Flow paths included in the first piping channel, the second piping channel, and the third piping channel are not limited to the flow paths described above, and may be combinations of any flow paths in the beverage filling machine **20**. In addition to the first piping channel, the second piping channel, and the third piping channel, one or more other piping channels that differ from these may be provided. In this case, in addition to the first CIP processing, the second CIP processing, and the third CIP processing, one or more CIP processing may be performed. It is preferable that all flow paths in the beverage filling machine **20** be included in at least one of a plurality of piping channels including the first piping channel, the second piping channel, and the third piping channel.

(92) It is preferable that the flow rate of the CIP cleaning liquid that flows through the first piping channel, the second piping channel, and the third piping channel be appropriately set based on the ability of pumps, the diameter of each piping, and the like. To be specific, among the first piping channel, the second piping channel, and the third piping channel, the flow rate (L/min) of the CIP cleaning liquid that flows through one of the piping channels in which the flow rate is the smallest may be 10% or more and preferably 20% or more of the flow rate (L/min) of the CIP cleaning liquid that flows through another of the piping channels in which the flow rate is the largest. Among the first piping channel, the second piping channel, and the third piping channel, the flow rate (L/min) of the CIP cleaning liquid that flows through one of the piping channels in which the flow rate is the smallest is 100% or less and may be 90% or less of the flow rate (L/min) of the CIP cleaning liquid that flows through another of the piping channels in which the flow rate is the largest.

(93) While sequentially CIP performing CIP processing on the inside of the beverage filling machine **20** in this way, the controller **60** monitors the temperature T_a of the inlet side and the temperature T_b of the outlet side of the beverage filling machine **20**. To be specific, the temperature T_a of the inlet side of the beverage filling machine **20** may be monitored by using the thermometer **68b**. The temperature T_b of the outlet side may be monitored by using the thermometer **68c**. As described above, during CIP processing, only one of the first piping channel, the second piping channel, and the third piping channel is performed CIP processing and the other piping channels are not performed CIP processing. That is, during CIP processing, a pipe line in which the CIP

cleaning liquid does not flow is present in the beverage filling machine **20**. Regarding this, during CIP processing, the controller **60** determines that, as long as the temperatures T_a and T_b each maintain a predetermined threshold temperature or higher, a pipe line through which the CIP cleaning liquid does not flow at the time also maintains sterility and can continue CIP processing. If the pipe line is opened, as long as sterilization of the inside of the aseptic chamber **13** has been finished, even in a case where the temperature of the pipe line in which the CIP cleaning liquid does not flow decreases and the pressure of the inside of piping becomes a negative pressure, it can be said that theoretically the pipe can be maintained in an aseptic state. That is, even if a pipe line in which the CIP cleaning liquid does not flow and whose temperature is lower than the threshold temperature is present in the beverage filling machine **20**, the pipe line is not opened to the atmosphere in a non-aseptic state. Therefore, entry of bacteria into the pipe line is suppressed, and it can be determined that an aseptic state is maintained. The controller **60** may stop CIP processing if the temperatures T_a or T_b becomes lower than the predetermined threshold temperature during CIP processing. The threshold temperature may be a predetermined temperature that is 85° C. or higher and lower than 100° C., and may be, for example, 90°. The CIP cleaning liquid may release heat by passing through the secondary (downstream) side of the filling nozzle **72** (the snift line **78**, the snift manifold **56**, the CIP manifold **59**, and the CIP circulation piping **81**), and the temperature T_b of the outlet side of the beverage filling machine **20** may become lower than the predetermined threshold temperature. In this case, the temperature of the CIP cleaning liquid measured by using a thermometer **68e** (see FIG. 2), which is set in each filling nozzle **72**, may be used instead of the temperature T_b of the outlet side. To be specific, the lowest of the temperatures of the CIP cleaning liquid measured by using the thermometers **68e**, which are set in all of the filling nozzles **72**, may be used as the temperature T_b . The setting position of the thermometer **68e** is not limited to the filling nozzle **72**, and the thermometer **68e** may be set in at least one of the snift line **78**, the snift manifold **56**, the CIP manifold **59**, and the CIP circulation piping **81**.

(94) In this way, according to the present embodiment, the first CIP processing (FIG. 5) for performing CIP processing on the first piping channel including the beverage supply line **73**, the second CIP processing (FIG. 6) for performing CIP processing on the second piping channel including the counter gas line **74**, and the third CIP processing (FIG. 7) for performing CIP processing on the third piping channel including the snift line **78** are performed. Thus, the beverage filling machine **20** for a carbonated beverage can be performed CIP processing efficiently and speedily. That is, in the beverage filling machine **20** for a carbonated beverage, many flow paths are present in the vicinity of the filling nozzle **72** compared with a filling nozzle for a non-carbonated beverage and have a complex structure. Therefore, it is difficult to fill all flow paths that are present in the vicinity of the filling nozzle **72** simultaneously with the CIP cleaning liquid. This is because, if all flow paths that are present in the vicinity of the filling nozzle **72** are attempted to be filled with the CIP cleaning liquid, the abilities of the pumps **51**, **52**, **91**, and the like that are present in the beverage supply piping **65** and the CIP circulation piping **81** may be insufficient or may be affected by the pressure loss of each piping. In this case, although it may be considered to convert facilities by, for example, reinforcing the pumps, increasing the diameter of piping, increasing the size of the valves, these measures are not realistic in view of cost and the like. According to the present embodiment, during CIP processing, piping in the beverage filling machine **20** is divided into a plurality of piping channels, and these piping channels are sequentially performed CIP processing. Thus, without considerably converting the beverage filling system **10**, it is possible to perform CIP processing on the beverage filling machine **20** for a carbonated beverage efficiently.

(95) In the above description, a case where CIP processing and SIP processing are simultaneously performed without additionally performing SIP processing has been described as an example (CSIP processing). This is not a limitation, and SIP processing may be performed after CIP processing. This SIP processing is processing for sterilizing the inside of a flow path through which a beverage is to pass before starting an operation of filling a container with the beverage. The SIP processing is

performed by, for example, causing heat steam or hot water to flow through a flow path that has been cleaned by the CIP cleaning. Thus, the inside of the flow path through which a beverage is to pass is sterilized to be in an aseptic state.

(96) In the above description, CSIP processing, with which cleaning and sterilization are simultaneously performed while circulating an alkaline cleaning liquid or an acid cleaning liquid, has been described as an example of CIP processing. However, in a rinsing step after circulating a CIP cleaning liquid, by supplying water whose sterility is ensured to the beverage supply piping **65**, SIP processing may be performed while washing away the CIP cleaning liquid with the water.

(97) According to the present embodiment, the heater **93** and the holding tube **62** are provided in the CIP circulation piping **81**, and a CIP cleaning liquid heated by the heater **93** is set in such a way as to pass through the holding tube **62** while spending a predetermined residence time or longer. Thus, it is possible to ensure the sterility of the CIP cleaning liquid and to supply the CIP cleaning liquid whose sterility is ensured to the beverage supply piping **65**.

(98) According to the present embodiment, the thermometer **68b** is provided on the outlet side of the holding tube **62** of the CIP circulation piping **81**, and the controller **60** monitors the F-value calculated based on the temperature of the thermometer **68b**. Thus, the controller **60** can determine that, as long as the F-value maintains a predetermined value or greater, the sterility of the CIP cleaning liquid that passes through the holding tube **62** is ensured. Moreover, by controlling the sterility of the CIP cleaning liquid by using the F-value, it is not necessary to perform CIP processing for an unnecessarily long time, and thus it is possible to reduce the product switching time of the beverage filling system **10** and to increase the production capacity.

(99) In the above description, as the beverage filling system, the beverage filling system **10** that uses an aseptic filling method has been described as an example. However, this is not a limitation. The beverage filling system may be, for example, a beverage filling system that uses a hot filling method for filling a container with a beverage under a high temperature of 55° C. or higher and 95° C. or lower. The present disclosure is applicable to any beverage filling system for which SIP processing (deactivation of microorganisms) is performed after CIP processing, such as a chilled beverage filling system, an alcohol beverage filling system, or the like.

(100) It is possible to appropriately combine a plurality of elements disclosed in the embodiment and modifications described above, as necessary. Alternatively, some elements may be omitted from all of the elements disclosed in the embodiment and modifications described above.

Claims

1. A beverage filling system for filling a container with a beverage, comprising: beverage supply piping that supplies the beverage; a beverage filling machine that is coupled to the beverage supply piping; a beverage sterilizing device that is coupled to the beverage supply piping; CIP circulation piping that couples the beverage filling machine with a midway of the beverage supply piping, that feeds a cleaning liquid flowed out from the beverage filling machine during CIP processing toward the beverage supply piping, and that circulates the cleaning liquid; and a controller that controls the beverage filling system, wherein a heater that heats the cleaning liquid that flows through the CIP circulation piping and a holding tube through which the cleaning liquid heated by the heater passes and includes a coil-shaped curved tube, a straight tube, or a spiral tube are provided in the CIP circulation piping and outside the beverage sterilizing device, and wherein the cleaning liquid heated by the heater is set in such a way as to pass through the holding tube while spending a predetermined residence time or longer before being supplied to the beverage supply piping.
2. The beverage filling system according to claim 1, wherein a thermometer is provided on an outlet side of the holding tube of the CIP circulation piping, and the controller monitors an F-value calculated based on a temperature of the thermometer.
3. The beverage filling system according to claim 2, wherein a time required by the cleaning liquid

to pass through the holding tube is predetermined based on an F-value required to sterilize the beverage.

4. The beverage filling system according to claim 1, wherein the cleaning liquid is heated to 85° C. or higher and lower than 150° C. in the heater.

5. The beverage filling system according to claim 1, wherein an aseptic tank is provided in the beverage supply piping, a bypass flow path that couples the CIP circulation piping on an outlet side of the holding tube and the beverage supply piping on an outlet side of the aseptic tank is provided, and the bypass flow path causes the cleaning liquid to flow from the holding tube side to the beverage supply piping on the outlet side of the aseptic tank without causing the cleaning liquid to pass through the aseptic tank.

6. A CIP processing method for performing CIP processing on a beverage filling system for filling a container with a beverage, the beverage filling system including beverage supply piping that supplies the beverage, a beverage filling machine that couples the beverage supply piping with a midway of the beverage supply piping, a beverage sterilizing device that is coupled to the beverage supply piping, and CIP circulation piping that is coupled to the beverage filling machine, that feeds a cleaning liquid flowed out from the beverage filling machine during CIP processing toward the beverage supply piping, and that circulates the cleaning liquid, the CIP processing method comprising: a step of heating the cleaning liquid that flows through the CIP circulation piping by using a heater; and a step of causing the cleaning liquid heated by the heater to pass through the holding tube which is provided in the CIP circulation piping and outside the beverage sterilizing device, and includes a coil-shaped curved tube, a straight tube, or a spiral tube, wherein the cleaning liquid heated by the heater is set in such a way as to pass through the holding tube while spending a predetermined residence time or longer before being supplied to the beverage supply piping.
